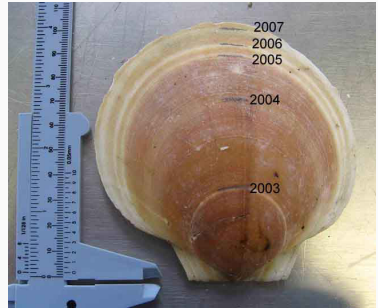


Sea Scallop Life History Parameters

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Sea Scallop Growth



Growth estimation based on shell growth rings. Because first ring(s) are sometimes obscure or missing, we treat the distance between the rings as an annual growth increment rather than attempting to estimate the shell's age.

We use a method introduced by Hart and Chute (ICESJMS 2009) to estimate the von Bertalanffy parameters L_∞ and K from shell growth increments using a mixed-effects linear regression:

$$r_2 = b + mr_1$$

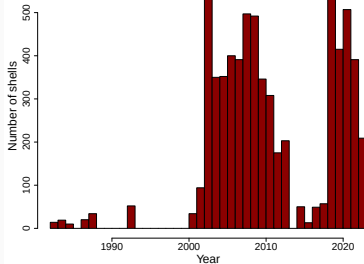
where r_1 and r_2 are the starting and ending shell height of the increment. There are individual-level random effects on both the intercept and slope, and station-level random effects on the intercept only.

The estimates of m , b and their variance among individuals can be used to estimate L_∞ , K and their variance among individuals, which in turn is used to estimate stochastic growth matrices.

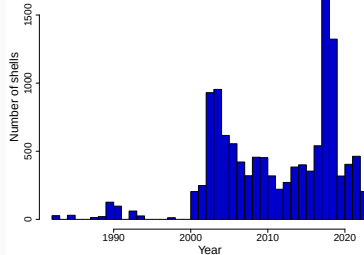
Shells by year

Limited shells analyzed from 1982–2000, with better coverage in recent years

Mid-Atlantic



Georges Bank



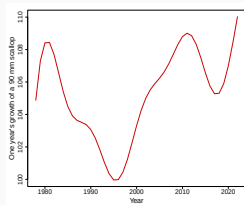
Exploring year effects

Mixed-effects GAMs were used to estimate growth variability by year:

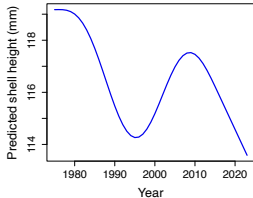
$$r_2 \sim r_1 + s(D, L) + s(y) \quad (1)$$

where $s(D, L)$ is a two dimensional smoother on depth and latitude, and $s(y)$ is a smoother on year. This was used to predict variability of growth by year, at fixed shell height, depth, and latitude.

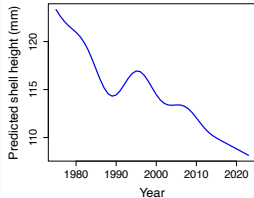
Mid-Atlantic



GB Closed



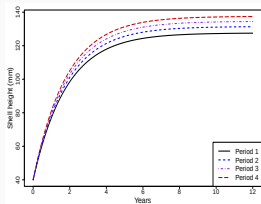
GB Open



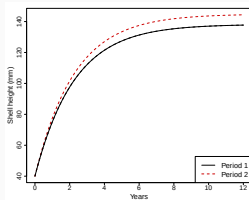
Growth Periods

Based on the GAMs, we split the time series into growth periods used in the CASA models, for which we calculate separate von B parameters and stochastic growth matrices: Four growth periods in the Mid-Atlantic, and two each for Georges Bank Open and Closed.

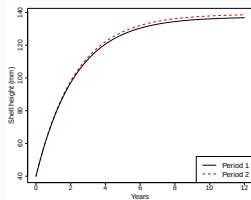
Mid-Atlantic



GB Closed



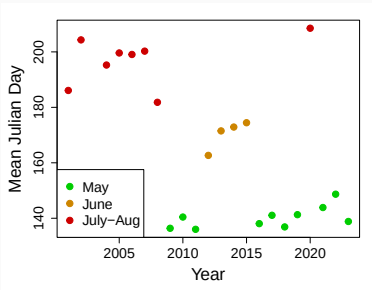
GB Open



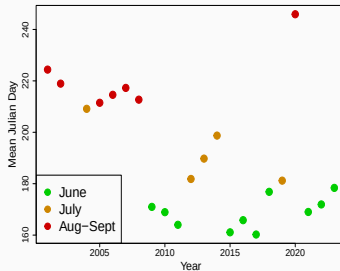
Shell height to meat weight relationships from survey data

Shell height to meat weight data has been collected on dredge surveys since 2001. Meat weights vary seasonally, so variations in survey timing need to be accounted for.

Mid-Atlantic



Georges Bank



Mixed-effects GAM models were used to account for survey timing, both without and with spatial covariates:

$$W = \ln(H) + s(J, Y) \quad (\text{MA - No covariates})$$

$$W = D \ln(H) + L + s(J, Y) \quad (\text{MA - with covariates})$$

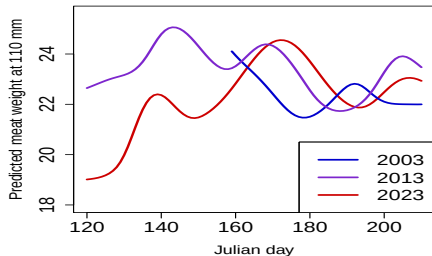
$$W = \ln(H) + s(J) + \text{as.factor}(Y) \quad (\text{GB - no covariates})$$

$$W = D \ln(H) + \text{as.factor}(Y) + L + s(J) \quad (\text{GB - with covariates})$$

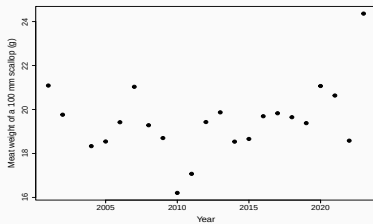
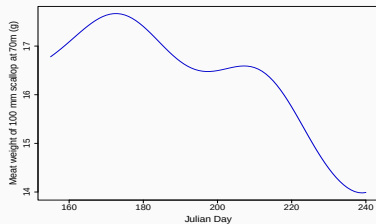
where W is meat weight, H is shell height, J is Julian day, Y is year, and L is latitude, using a gamma distribution with a log link; station-level random effects on the intercept only.

Year and Day effects

Mid-Atlantic



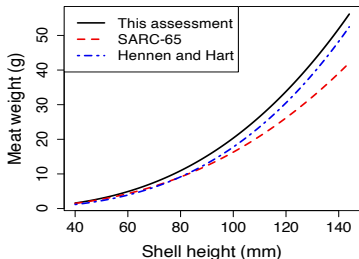
Georges Bank



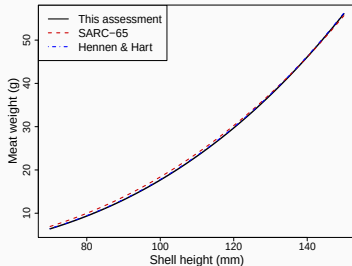
Historical comparisons

Baseline shell height to meat weight relationships were predicted using the year 2010 and Julian day 150 in the Mid-Atlantic and 180 on Georges Bank.

Mid-Atlantic



Georges Bank



Commercial meat weights from at-sea observers

Observers collected about 100 retained scallops, measure their shell heights, and then obtain the aggregate volume of their meats. This is converted to weight, and a predicted meat weight is calculated from the survey sh to mw relationships, and a meat weight anomaly is computed:

$$A = \frac{\text{predicted} - \text{observed}}{\text{predicted}}. \quad (2)$$

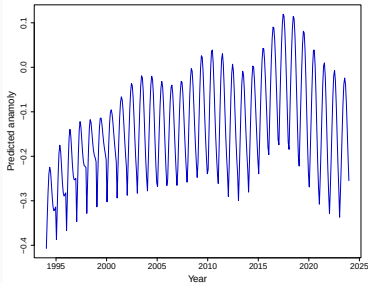
The mean meat weight anomaly was calculated for each month and year, which was then fit by a GAM:

$$\text{MWanomaly} \sim s(\text{Year}, \text{Month}) + \text{nscallop} + \text{Clop}, \quad (3)$$

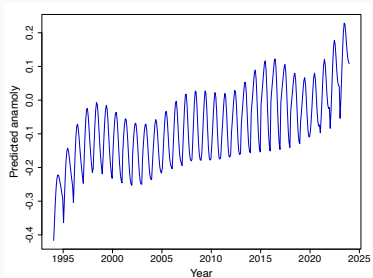
with the Gaussian family and identity link.

Meat weight anomalies

Mid-Atlantic



Georges Bank



Annual anomalies are computed as the weighted (by landings) mean of the monthly anomalies. When calculating landings by weight in the CASA model, landings at shell height are converted into weights by the survey SHMW relationship, and then adjusted by the annual anomaly. SYM and SAMS use the mean weighted anomalies from the last five years.

Natural mortality

Estimates of natural mortality M can be made based on longevity, i.e., the age of the oldest observed animal $t_{\max}$. Here we will apply three such estimators:

$$M = \exp(1.23 - 0.834 \ln t_{\max}) \quad (\text{Hoenig 1983})$$

$$M = 4.899 t_{\max}^{-0.916} \quad (\text{Then et al. 2015})$$

$$M = 5.4 / t_{\max} \quad (\text{Hamel and Cope 2022})$$

We used the oldest aged shell to estimate $t_{\max}$, assuming that the first growth ring was age 2. This is likely correct for the oldest shells in the two regions.

Natural mortality estimates

Mid-Atlantic: Oldest shell is 14 years old. The three estimators give $M = 0.38$, 0.44 and 0.39 , respectively. The working group concluded that $M = 0.4$ is a reasonable estimate for the Mid-Atlantic. This is considerably higher than the $M = 0.25$ estimate used in recent past assessments.

Georges Bank: The oldest shell was 19 years old, which gives M estimates of 0.29 , 0.33 , and 0.28 , respectively. The Georges Bank Closed CASA model estimated a mean $M = 0.27$. We used $M = 0.27$ in the Georges Bank Open CASA model and Georges Bank SYM model. This is higher than the estimate of $M = 0.2$ in previous recent assessments.